Personal Time Tracker: Requirements

For Manav · Draft v1 · 7 November 2026

Items marked *(assumption)* were not stated directly and should be confirmed before building.

Goal

One system that tracks study, music and content in the way you already work: hours, focus, topics and weekly growth. It shows graphs (hourly, daily, weekly), keeps your study bank, flags gaps in your log and gives you tips to be more focused.

First feature: import from the WhatsApp log

- You import your exported WhatsApp chat (the .zip or .txt) first, so your history from 13 October 2026 appears on day one.
- The importer reads each timestamp and turns messages into events: login, logout, on desk, off desk, breakfast, lunch, break and riyaz on/off.
- Your messy wording ("Off", "out", "OTD", "bon", "Luck off") is matched using a word list that you can edit.
- Random notes that are not time entries, such as numbers and amounts, are ignored or put in a review list. They are never guessed.
- Missing or unclear entries, such as an "on" with no "off", go to the "to fill" list.
- Before anything is saved, you see a preview of what was understood. *(assumption)*
- Importing again later must not create duplicates. *(assumption)*

What you log

- **Study:** start and stop, with the topic (for example `S on Compiler Design` and `S off`). All logged study time counts as quality.
- **Break:** start and stop, taken from your 2h break bank.
- **Music (riyaz):** a "done today" tick, with hours optional and no timer.
- **Content:** start and stop, like study. *(assumption)*
- **Day start and end:** your Login and Logout. *(assumption)*

How you log

- You type short codes or tap buttons, whichever is faster.
- It is laptop-first and works in the browser at your desk.

Bank rules

- Each day gives you 6h of study and 2h of break.
- A study shortfall and any unused break time carry forward to the next day.
- Saturday and Sunday are full days off with no quota. The bank resets at the start of Saturday. *(assumption)*

Gaps

- Gaps in the log and missing "off" entries are flagged on the timeline and collected in a "to fill" list.
- You fill them in whenever you like.

Dashboard

- **Game side:** XP, levels, streaks and your bank balance.
- **Analytics side:** a daily timeline, weekly totals, an hourly heatmap (hour by weekday) and hours per topic.

Insights and growth

- For now, growth means hours per topic plus weekly results. You can type the results, or the system can work them out from your topic logs.
- You get a weekly summary and daily tips.
- Tips about your own patterns, such as your best focus hours, come from your log. They need about two weeks of data before they mean much.
- General study-science tips, such as retrieval practice and spaced repetition, are marked as general so they stay separate from your own data.

Still open

- What earns XP and levels, for example hitting the 6h quota or a streak of days.
- Where the data lives. The suggested start is your laptop's browser with an export button, and later it can move into your habit tracker app.

Suggested build order

1. WhatsApp import, the word list and the review list.
2. Dashboard with charts, the bank and gap flags.
3. Quick logging by typing and tapping.
4. Insights, tips and XP.